A Conley–Morse Framework for Neural Population Hybrid Dynamics Analysis

(Hybrifolds, Conley-More Indexes and Cognitive-Dynamic Symbols)

Oct 21, 2026 · @Angus

1. Single-region pipeline

Describe a session's dynamics as a hierarchy of labelled DAGs: nodes are metastable sets (the formal version of brain states/regimes), edges are directed transitions, labels are Conley indices plus descriptive statistics. The decomposition/reconstruction logic follows Conley theory as surveyed by Kokubu & Spendlove; the empirical motivation that decision dynamics contain regime transitions recoverable from spiking comes from Luo et al. (2025).

Steps, with one choice made at each:

1. **Preprocess and embed — chosen once.** Bin (10–25 ms), √-transform, smooth, soft-normalize; one dimensionality-reduction method (e.g., PCA with delay coordinates), fixed for the whole analysis. The theory is coordinate-agnostic; this choice is ours and is reported as such.
   NOTE (Hybrifolds not manifolds): The regular embedding methods like MARBLE or CEBRA, simply cannot handle reset maps or sudden jumps. Their methods actively hide them. See the **Hybrid dynamic embedding and the reset maps section**.
2. **Estimate the flow field — chosen once.** Either empirical displacement averages over visits, or a fitted field via FINDR (Kim et al., 2023). One method, used throughout.
3. **Flow → DAG at scale (ε, τ).** Cover the embedded space with boxes, build the box map, take SCCs (Morse sets) and their condensation (the DAG), per the database methodology (Arai et al., 2009).
4. **Sweep scale → hierarchy.** Repeat step 3 over a ladder of box sizes; track nodes by overlap. Birth–death of nodes across resolution is the scale barcode, and the surviving tree of symbols is the backbone used by every later section (Dey, Lipiński, Mrozek, & Slechta, 2022).
5. **Label nodes.** Conley index per node (stability type, unstable dimension; certificates of fixed point or periodic orbit), plus attributes: dwell times, occupancy, contraction rates, local dimension, radius — intrinsic and embedding-dependent attributes stored separately, in the spirit of the capacity-linked effective geometry of Chou et al. (2025).
6. **Recursive atlas (the novel step).** Within each node, re-fit the *same* embedding method to that node's timepoints, re-estimate the flow, compute child DAGs; iterate until symbol boundaries stabilize under a prediction-error or description-length criterion. Benchmark against rSLDS (Linderman et al., 2017).

7. **Trial structure as hybrid dynamics.** Within-trial flow plus a between-trial reset map; the across-trial objects (history bias, block prior) are periodic orbits of the reset system, analyzed through the hybrid suspension and hybrid index pairs of Rivas & Kalies (2026), whose cubic reset example (bistable perseveration, $S_0 \to S_1$, S_2) is the template. Candidate method: box-map edges whose length does not shrink as $\varepsilon \to 0$ as a data-driven detector of resets.
8. **Change across learning or conditions.** Track index generators over a parameter axis (session, delay, block) with the Conley–Morse persistence barcode (Dey, Lipiński, & Soriano-Trigueros, 2026): bar births/splits/annihilations = landscape construction; intact bars with drifting attributes = refinement within it.

Stage	Supplied by	Output	Corpus reference
Coordinates	One embedding, chosen once (PCA + delays / CEBRA) — outside the theory	Low-D trajectory	—
Arrows	One flow estimator (empirical averages or FINDR) — outside the theory	Vector field / box map	Kim et al. (2023); Luo et al. (2025)
Nodes + DAG	SCC/condensation at scale (ε, τ) (CMGDB)	Symbols + order	Arai et al. (2009); Kokubu & Spendlove (n.d.)
Node label	Cubical relative homology (CHomP)	Conley index	Kokubu & Spendlove (n.d.)
Node attributes	Descriptive statistics per node	Attribute dictionary	Chou et al. (2025)
Hierarchy	ε/τ sweep	Scale barcode + symbol tree	Dey et al. (2022)

Trial resets	Hybrid suspension, hybrid index pairs	Across-trial Morse graph	Rivas & Kalies (2026)
Change tracking	Zigzag persistence of index generators	λ-barcode	Dey et al. (2026)
Atlas refinement	Recursion (ours)	Per-node embeddings, neuron loadings	— (novel)

Hybrid dynamic embedding: unsolved problem

The hybrid Conley machinery of Rivas & Kalies (2026) analyzes a hybrid system whose guard G and reset map r are given; it does not identify them from data. The identification problem splits into three cases.

1. **Guard known (task settings): solved.** Trial boundaries give G; the (pre-reset, post-reset) state pairs across trials are samples of r, fit by regression. The fitted map's fixed-point structure (one vs. three) is then read off directly.
2. **Guard unknown: open for neural data.** Control theory treats this as hybrid/switched-system identification. Hybrid-SINDy recovers piecewise models and their switching surfaces by sparse regression with information criteria (Mangan, Askham, Brunton, Kutz, & Proctor, 2019). The same group behind the hybrid Conley index has paired convex-optimization identification of switching systems with combinatorial outer approximations, yielding certified Morse graphs and regions of attraction from trajectory data (Rivas, Kalies, Iwasaki, Bloch, & Ghaffari, 2025) — the identification half our pipeline's step 7 needs, demonstrated so far on planar examples. CHyLL instead learns a latent embedding in which the hybrid flow becomes continuous — a learned analogue of the suspension construction — with no event detection or per-mode models (Teng, Liu, Song, & Sreenath, 2025).
3. **A warning for embedded data.** Delay embedding assumes a diffeomorphism; a reset violates it, and delay vectors spanning the jump smear the discontinuity into a spurious fast arc connecting the two regions. Smoothing does the same. A naive pipeline therefore converts resets into fake fast flows silently. Candidate detector (ours, untested): box-map transitions whose length does not shrink as $\varepsilon, \tau \to 0$; flow edges contract with resolution, reset edges do not. The sources of persistent long edges estimate G, their targets estimate r(G), reducing case 2 to case 1.

The hybrifold perspective, and candidate solutions. The geometric object underneath is the *hybrifold*: the modes' domains glued along the guard by the reset map, so the hybrid flow becomes continuous on a single piecewise-smooth space — introduced by Simić et al. (2005), "Towards a geometric theory of hybrid systems." The Rivas & Kalies (2026) suspension is its semiflow-friendly variant, stretching the jump over unit time instead of gluing instantaneously. The embedding problem is therefore: learn a hybrifold-like space from data, rather than forcing a smooth embedding onto dynamics that contain seams. Candidate solutions:

- Mangan et al. (2019), "Model selection for hybrid dynamical systems via sparse regression" (Hybrid-SINDy): clusters local models, fits sparse per-mode equations, and selects switching surfaces by information criteria — coordinates stay fixed, modes and guards are learned.
- Rivas et al. (2025), "Topological dynamics via learned hybrid systems": convex-optimization identification of switching systems paired with combinatorial outer approximations, yielding certified Morse graphs and regions of attraction — the identification companion to the hybrid Conley index, demonstrated so far on planar examples.
- Teng et al. (2025), "CHyLL: Learning continuous neural representations of hybrid systems": learns a latent space in which the hybrid flow is continuous — in effect a learned hybrifold/suspension, needing no event detection or per-mode models, and a candidate replacement for the hand-chosen embedding of step 1 when resets are present.

Unstated experimental evidence for mid-trial resets. Luo et al. (2025) report an abrupt, coordinated change in both dynamical regime and neural mode at decision commitment: the activity's direction of change rotates to a nearly orthogonal mode, the timing (nTc) varies trial to trial and is locked to neither stimulus nor response, and evidence stops influencing choice after it. They do not frame this as hybrid dynamics, but it is the signature a state-dependent guard would produce — a reset or jump *inside* the trial, triggered by where the trajectory is rather than by the clock. Reset identification therefore cannot assume guards sit only at trial boundaries; commitment-like internal guards must be discoverable too, which is exactly what the unsupervised detector above is for.

2. Across regions: communication via symbols

Each region is solved separately by Section 1; region B's dynamics are then treated as parameterized by region A's symbol (a skew-product). Communication separates into two measurable forms.

Within-regime coupling: reduced-rank regressions conditioned on the joint symbol, giving a communication subspace per DAG state rather than one global subspace (extending Semedo et al., 2019).

Regime-setting coupling: A's symbol as the parameter axis of B's persistence barcode, stating which of B's attractors are created, split, or annihilated by A's state. The local object is the attractor–repeller split (AR-split) diagram (Dey et al., 2026): it records how one invariant set decomposes across the change and which Conley index generators pass to each piece, and the transition diagram fusing the AR-splits yields the barcode. The combinatorial multivector-field framework of the same paper is the natural formalism, being finite from the outset.

A symbolic prediction test (does A's symbol improve prediction of B's transitions on B's DAG scaffold, beyond B's own history) gives directionality. An occasional joint fit checks for states invisible in the marginals. Across animals, where no parameter path connects systems, attributed DAGs are compared with Fused Gromov–Wasserstein distances.

Stage	Supplied by	Output	Corpus reference
Within-regime coupling	Reduced-rank regression per joint symbol	Communication subspace per DAG state	Semedo et al. (2019)
Regime-setting coupling	AR-split diagrams + B's λ-barcode, with A's symbol as the parameter axis	Map of which B-attractors A's state creates/splits/annihilates, and who inherits which generators	Dey et al. (2026)
Directionality	Transition models on B's DAG scaffold (Granger-style, symbol level)	Directed influence estimates	Comparison class: Ashwood et al. (2022)
Joint check	Occasional joint-space Morse decomposition	Joint DAG vs. product of marginals	—
Cross-animal comparison	FGW on attributed DAGs	Distances between landscapes	Vayer et al. (2019)

3. Back to neurons: hierarchical GLMs

This is the least specified part of the program; the exact model form cannot be fixed a priori and will require tinkering against data.

The organizing idea: the symbol tree from Section 1 supplies the *hierarchy* for the model, so symbolic covariates enter as **hierarchical effects**, not flat dummy variables. Each neuron's response to a fine symbol is modelled as a deviation from its response to the parent coarse symbol, with partial pooling down the tree — so coarse, robust states absorb the main effects and fine, short-lived symbols only earn separate coefficients where the data demand them. Kernels likewise attach at levels of the hierarchy: transitions exist at every scale, and a transition between coarse states and a transition between that state's children are different events, modelled at their own level. What the persistence structure supplies, then, is the *nesting* — which effects group under which — rather than fixed timings and widths.

Candidate implementations to be compared empirically: random effects nested along the symbol tree; tree-structured group-lasso penalties; or pooled kernels with per-level deviations. Covariates in any version: own-region symbol (hierarchical), other-region symbol (hierarchical), continuous position (distance to saddle, depth in basin), transition-locked kernels. The per-node embedding loadings from the recursion already give each neuron a profile across symbols; the GLM is the out-of-sample test of that assignment, with DAGs estimated on one half of trials and models fit on the other. The comparison class is the GLM-HMM (Ashwood et al., 2022); the question that closes the program is which cell-level properties (type, channel expression, connectivity) predict a neuron's placement in the landscape. The symbolic covariates themselves derive from the barcode and its AR-split diagrams (Dey et al., 2026): the discretization is what makes the geometry interpretable, and the GLM demonstrates that the same discretization is predictive — that the symbols describe individual neurons en masse.

The emergence test (the author's proposal). Hold out a small population of neurons within the region; compute the embedding, flow field, and symbols from the remaining units only; then test whether the symbols predict the held-out units' spiking out of sample. This is a direct test of emergence rather than a circular fit: if the population geometry is a real collective property, enough neurons should suffice to recover it, and units that never contributed to its construction should nonetheless be described by it. (The circularity concern this answers — that symbols fit and tested on the same units are partly guaranteed to predict them — was raised by Claude in discussion.) Two strengthenings: the symbol should add predictive value beyond the continuous latent position, so the boundary itself is shown to be real; and symbol identity should predict changes in the held-out units' own dynamics — autocorrelation timescale, burstiness, variance — not only mean rate.

Hypothesis: composition by timescale (the author). If intrinsic timescales gate which dynamics a neuron can participate in, the manifold is not expressed by all neurons but *built from the subset whose timescales match it*. Predictions: (i) held-out prediction quality is not uniform — it correlates with the unit's intrinsic timescale relative to the symbol's dwell time, so a slow symbol predicted well by slow units and poorly by fast ones is evidence for composition, not a

failed emergence test; (ii) per-symbol loadings regressed on intrinsic timescale distinguish *modulation* (all units ride the manifold with varying gain) from *selection* (a timescale-matched subpopulation carries it); (iii) because symbols at different levels of the hierarchy have different dwell times, the full prediction is a matching matrix — coarse, long-lived symbols carried by slow units, fine fast symbols by fast units — making the symbol tree a timescale decomposition of the population. Intrinsic timescale is measurable per unit and partly set by cell type and channel expression, so this hypothesis is the bridge from the geometry back to biophysics.

Stage	Supplied by	Output	Corpus reference
Model class	Regularized GLM per unit	Spike predictions	Comparison class: Ashwood et al. (2022)
Effect structure	Symbol tree from scale barcode → nested/partially pooled effects	Hierarchical symbol coefficients	Dey et al. (2022) (supplies the tree)
Symbol covariates	Barcode generators + AR-split events (discretized geometry)	Interpretable covariates with predictive value	Dey et al. (2026)
Kernels	Transition-locked, attached per level of the tree — exact form open	Multi-scale temporal responses	— (open; requires tinkering)
Continuous covariates	Geometry within nodes (distance to saddle, basin depth)	Mechanistic position effects	Rivas & Kalies (2026); Luo et al. (2025)
Neuron–symbol assignment	Loadings from the recursive atlas (Section 1, step 6)	Per-neuron landscape profile	— (novel)

Validation	Held-out-neuron emergence test (author) + split-trial estimation vs. fitting	Out-of-sample test of the whole stack on single units	— (novel)

References

Arai, Z., Kalies, W., Kokubu, H., Mischaikow, K., Oka, H., & Pilarczyk, P. (2009). A database schema for the analysis of global dynamics of multiparameter systems. *SIAM Journal on Applied Dynamical Systems, 8*(3), 757–789.

Ashwood, Z. C., Roy, N. A., Stone, I. R., International Brain Laboratory, Urai, A. E., Churchland, A. K., Pouget, A., & Pillow, J. W. (2022). Mice alternate between discrete strategies during perceptual decision-making. *Nature Neuroscience, 25*(2), 201–212.

Chou, C.-N., Kim, R., Arend, L. A., Yang, Y.-Y., Mensh, B. D., Shim, W. M., Perich, M. G., & Chung, S. (2025). Geometry linked to untangling efficiency reveals structure and computation in neural populations. *bioRxiv*.

Dey, T. K., Lipiński, M., Mrozek, M., & Slechta, R. (2022). Persistence of Conley–Morse graphs in combinatorial dynamical systems. *SIAM Journal on Applied Dynamical Systems, 21*(2), 817–839.

Dey, T. K., Lipiński, M., & Soriano-Trigueros, M. (2026). Conley–Morse persistence barcode: A homological signature of combinatorial bifurcations. *Foundations of Computational Mathematics, 1–59*.

Gallier, J., & Quaintance, J. (2020). *Differential geometry and Lie groups: A computational perspective*. Springer.

Kim, T. D., Luo, T. Z., Pillow, J. W., & Brody, C. D. (2023). Flow-field inference from neural data using deep recurrent networks. *bioRxiv*.

Kokubu, H., & Spendlove, K. (n.d.). *A user's guide to the Conley–Morse database* [Survey manuscript].

Linderman, S., Johnson, M., Miller, A., Adams, R., Blei, D., & Paninski, L. (2017). Bayesian learning and inference in recurrent switching linear dynamical systems. *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics (AISTATS), PMLR 54*, 914–922.

Luo, T. Z., Kim, T. D., Gupta, D., Bondy, A. G., Kopec, C. D., Elliott, V. A., ... & Brody, C. D. (2025). Transitions in dynamical regime and neural mode during perceptual decisions. *Nature, 646*(8087), 1156–1166.

Mangan, N. M., Askham, T., Brunton, S. L., Kutz, J. N., & Proctor, J. L. (2019). Model selection for hybrid dynamical systems via sparse regression. *Proceedings of the Royal Society A, 475*(2223), 20180534.

Rivas, B., & Kalies, W. (2026). Conley index theory for hybrid systems. *arXiv*. arXiv:2606.18501

Rivas, B., Kalies, W., Iwasaki, K., Bloch, A., & Ghaffari, M. (2025). Topological dynamics via learned hybrid systems. *arXiv*. arXiv:2511.08737

Semedo, J. D., Zandvakili, A., Machens, C. K., Yu, B. M., & Kohn, A. (2019). Cortical areas interact through a communication subspace. *Neuron, 102*(1), 249–259.

Simić, S. N., Johansson, K. H., Lygeros, J., & Sastry, S. (2005). Towards a geometric theory of hybrid systems. *Dynamics of Continuous, Discrete and Impulsive Systems, Series B, 12*(5–6), 649–687.

Teng, S., Liu, H., Song, J., & Sreenath, K. (2025). CHyLL: Learning continuous neural representations of hybrid systems. *arXiv*. arXiv:2512.10117

Vayer, T., Chapel, L., Flamary, R., Tavenard, R., & Courty, N. (2019). Optimal transport for structured data with application on graphs. *Proceedings of the 36th International Conference on Machine Learning (ICML), PMLR 97*, 6275–6284.

To verify before circulation: the Dey et al. (2022) volume/pages, the publication status of the Kokubu–Spendlove survey, and the Simić et al. (2005) journal volume/pages.